Hope Enterprise University College	
Faculty of Informatics	
Computer Science Department	
Course Outline	
Course Title:	**Compiler Design**
Course Code:	**CS 431**
Credit Hours:	**3**
Contact Hours:	**4**
Lab Hours:	**2**
Prerequisite:	**Automata and Complexity Theory**
Course Category:	**Major**
Year:	**IV**
Semester:	**I**
Instructor	**Shewatatek Lemma**

Course Description

The course deals with the various concepts involved in compiler construction including scanning, parsing, code optimization and generation.

Course Objectives

On completion of this course students should be able to:

1. Lexical analysis and regular expressions;
2. Parsing (top down and bottom up) and the construction of parsers;
3. Syntax directed translation;
4. Type checking and type systems;
5. Intermediate code generation and representation;
6. Code generation and related issues such as memory management and register allocation
7. Techniques of code optimization
8. The various tools used for the construction of compilers

Course Contents	Weeks
Chapter 1 Introduction	
1.1 Analysis and synthesis in a compilation	**1**

1.2 Phases in a compilation 1.3 Compiler construction tools	
Chapter 2: Lexical analysis 2.1 Token, pattern, lexeme 2.2 Attributes of a token 2.3 Errors 2.4 Input buffering (Buffer pairs) 2.5 Specification using regular expressions 2.6 Recognizing tokens using transition diagrams	**2 & 3**
Chapter 3: Syntax analysis 3.1 Role of a parser 3.2 Syntax error handling 3.3 Top-down parsing 3.3.1 Recursive Descent Parsing 3.3.2 Implementing RDP using recursive procedures 3.3.3 Implementing RDP using non-recursive predictive parsing 3.3.4 LL(1) grammars 3.3.5 Error recovery in predictive parsing 3.4 Bottom-up parsing 3.4.1 Stack implementation of shift-reduce parsing 3.4.2 Conflict during shift reduce parsing 3.4.3 LR parsers 3.4.4 LR grammars 3.4.5 SLR parsing 3.4.6 LR parsing 3.4.7 LALR parsing	**4, 5, 6, 7 & 8**
Mid Exam	**9**

Chapter 4: Syntax directed translation 4.1Syntax directed definitions 4.2 Classes of non-circular attributed grammars 4.2.1 S-attributed grammars 4.2.2 L-attributed grammars	**10 & 11**
Chapter 5: Type checking 5.1 Type systems 5.2 Specifications of a type checker **5.3** Equivalence of types	**11 & 12**
Chapter 6: Intermediate code generation (4 hours/1 weeks) 6.1 Intermediate languages 6.2 Syntax directed translation into three address code	**13-14**
Chapter 7: Code generation (4 hours/1 weeks) 7.1 Issues in the design of a code generator 7.2 Memory management 7.3 Instruction selection 7.4 Register allocation	**15 - 16**

Teaching and Learning Method:

- Lecture
- Problem solving/inquiry
- Lab sessions

Assessment /Evaluation Scheme

Component	Weight	Coverage	Assessment Time		
Mid	**20%**	**Chapter 1, 2, & 3**	**Week 9**		
Class Work	5%	Chapter 3, 4, 5 & 6	Week 13		
Test	15%	Chapter 4, 5, & 6	Week 13		
Project	10%	Chapter 1-7	Week 15		
Final Exam	**50%**	**All Chapters**			
Total	100%				

Text Book

Compilers: Principles, techniques and tools by Alfred V. Aho, Ravi Sethi, Jeffrey D. Ullman
Reference Books
1. Compiler construction for digital computers: Gries, David (QA 76.5.G73) 2. Compilers: Muller, John (QA 76.76.C65.M84)